Zhanhong Cheng

Curriculum Vitae (July 15, 2026)

Email: zhanhong.cheng@ufl.edu
Website: chengzhanhong.github.io
GitHub: github.com/chengzhanhong
Google Scholar, ORCID, Web of Science

Education

McGill University — Montreal, Canada
Ph.D. in Civil Engineering — Jan 2019–Oct 2022

- Advisor: Prof. Lijun Sun (McGill, main) & Prof. Martin Trépanier (PolyMtl, co-supervisor)
- Dissertation: "Travel-Behavior-Based Inference and Forecasting Methods in Metro System"

Harbin Institute of Technology — China
M.S. in Transportation Engineering — Sep 2016–Jul 2018

- Advisor: Prof. Jia Yao
- Thesis: "An Analysis of Two Hybrid Route Choice Models in Stochastic Assignment Paradox"

Harbin Institute of Technology — China
B.Eng. in Traffic Engineering — Aug 2012–Jul 2016

Employment

Zhejiang University — Hangzhou, China
ZJU 100 Young Professor, College of Civil Engineering and Architecture — May, 2026–Current

University of Florida — Gainesville, USA
Postdoctoral Associate, Department of Urban and Regional Planning, Urban AI Lab — Aug, 2024–Jan, 2026

- Advisor: Prof. Shenhao Wang

McGill University — Montreal, Canada
Postdoctoral Researcher, Department of Civil Engineering — Aug, 2022–Aug, 2024

- Advisor: Prof. Lijun Sun

Professional Experience

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea
Visiting Scholar, Transport and Urban Planning Arena (TUPA) — Mar 2026–May 2026

ExPretio — Montreal, Canada
Researcher — Sep 2022–Aug 2024

exo (Réseau de transport métropolitain) — Montreal, Canada
Intern — Feb 2019–Feb 2022

Journal Publications[1]

[1] **Z. Cheng**, L. Hu, Y. Bu, Y. Zhou, and S. Wang*, "Graph neural networks for residential location choice: Connection to classical logit models", *Transportation Research Part B: Methodological*, vol. 209, p. 103 464, 2026. DOI: 10.1016/j.trb.2026.103464

[2] X. Zhu, **Z. Cheng***, L. Miranda-Moreno, and L. Sun, "ProST-CP: Probabilistic spatiotemporal CP decomposition for bike-sharing origin-destination demand modeling", *Transportation Research Part C: Emerging Technologies*, 2026, In publishing.

[3] X. Chen, **Z. Cheng**, A. M. Schmidt, and L. Sun*, "Conditional forecasting of bus travel time and passenger occupancy with bayesian markov regime-switching vector autoregression", *Transportation Research Part B: Methodological*, vol. 192, p. 103 147, 2025. DOI: 10.1016/j.trb.2024.103147

[4] X. Chen, **Z. Cheng**, and L. Sun*, "Bayesian inference of time-varying origin–destination matrices from boarding and alighting counts for transit services", *Transportation Research Part B: Methodological*, vol. 199, p. 103 278, 2025. DOI: 10.1016/j.trb.2025.103278

[5] **Z. Cheng**, J. Wang, M. Trépanier, and L. Sun*, "Abnormal metro passenger demand is predictable from alighting and boarding correlation", *Transportation Research Part C: Emerging Technologies*, vol. 178, p. 105 239, 2025. DOI: 10.1016/j.trc.2025.105239

[6] **Z. Cheng***, V. J. Rodrigues de Sousa, K. Pérez Martínez, T. Barbier, and L. Sun, "Railway booking demand forecasting for revenue management: A deep probabilistic approach", *Transportmetrica A: Transport Science*, 2025. DOI: 10.1080/23249935.2025.2563182

[7] X. Chen, **Z. Cheng**, and L. Sun*, "Bayesian inference for link travel time correlation of a bus route", *Transportmetrica B: Transport Dynamics*, vol. 12, no. 1, p. 2 416 181, 2024. DOI: 10.1080/21680566.2024.2416181

[8] X. Chen, **Z. Cheng**, H. Cai, N. Saunier*, and L. Sun, "Laplacian convolutional representation for traffic time series imputation", *IEEE Transactions on Knowledge and Data Engineering*, 2024. DOI: 10.1109/TKDE.2024.3419698

[9] F. Wu, **Z. Cheng**, H. Chen, Z. Qiu*, and L. Sun, "Traffic state estimation from vehicle trajectories with anisotropic gaussian processes", *Transportation Research Part C: Emerging Technologies*, vol. 163, p. 104 646, 2024. DOI: 10.1016/j.trc.2024.104646

[10] Y. Zhao, Z. Ma, H. Peng, and **Z. Cheng***, "Predicting metro incident duration using structured data and unstructured text logs", *Transportmetrica A: Transport Science*, pp. 1–29, 2024. DOI: 10.1080/23249935.2024.2396951

[11] H. Cai, F. Wu, **Z. Cheng**, B. Li, and J. Wang*, "A large-scale empirical study on impacting factors of taxi charging station utilization", *Transportation Research Part D: Transport and Environment*, vol. 118, p. 103 687, 2023. DOI: 10.1016/j.trd.2023.103687

[12] X. Chen, **Z. Cheng**, J. Jin, M. Trépanier, and L. Sun*, "Probabilistic forecasting of bus travel time with a bayesian gaussian mixture model", *Transportation Science*, 2023. DOI: 10.1287/trsc.2022.0214

[13] X. Chen, C. Zhang, **Z. Cheng**, Y. Hou, and L. Sun*, "A bayesian gaussian mixture model for probabilistic modeling of car-following behaviors", *IEEE Transactions on Intelligent Transportation Systems*, 2023. DOI: 10.1109/TITS.2023.3334909

[14] **Z. Cheng**, M. Trépanier, and L. Sun*, "Real-time forecasting of metro origin-destination matrices with high-order weighted dynamic mode decomposition", *Transportation Science*, vol. 56, no. 4, pp. 904–918, 2022. DOI: 10.1287/trsc.2022.1128

[1] *: corresponding author; †: co-first author.

[15] **Z. Cheng**, X. Wang, X. Chen, M. Trépanier, and L. Sun*, "Bayesian calibration of traffic flow fundamental diagrams using gaussian processes", *IEEE Open Journal of Intelligent Transportation Systems*, 2022. DOI: 10.1109/OJITS.2022.3220926

[16] F. Wu, H. Chen, K. Hou, **Z. Cheng**, and T. Z. Qiu*, "Adaptive pushbutton control for signalized pedestrian midblock crossings", *Journal of Transportation Engineering, Part A: Systems*, vol. 148, no. 4, p. 04 022 011, 2022. DOI: 10.1061/JTEPBS.0000659

[17] J. Yao*, H. Yuan, Y. Jiang, S. An, and **Z. Cheng**, "Effect of route overlapping feature on stochastic assignment paradox", *Transportation Letters*, pp. 1–11, 2022. DOI: 10.1080/19427867.2022.2159058

[18] K. Zhu, **Z. Cheng**†, J. Wu, F. Yuan, and L. Sun*, "Quantifying out-of-station waiting time in oversaturated urban metro systems", *Communications in Transportation Research*, vol. 2, p. 100 052, 2022. DOI: 10.1016/j.commtr.2022.100052

[19] Z. Zhuang, **Z. Cheng**, J. Yao*, J. Wang, and S. An, "Bus travel time reliability incorporating stop waiting time and in-vehicle travel time with avl data", *International Journal of Coal Science & Technology*, vol. 9, no. 1, p. 71, 2022. DOI: 10.1007/s40789-022-00544-7

[20] **Z. Cheng**, M. Trépanier, and L. Sun*, "Incorporating travel behavior regularity into passenger flow forecasting", *Transportation Research Part C: Emerging Technologies*, vol. 128, p. 103 200, 2021. DOI: 10.1016/j.trc.2021.103200

[21] **Z. Cheng**, M. Trépanier, and L. Sun*, "Probabilistic model for destination inference and travel pattern mining from smart card data", *Transportation*, vol. 48, pp. 2035–2053, 2021. DOI: 10.1007/s11116-020-10120-0

[22] **Z. Cheng**, J. Yao*, A. Chen, and S. An, "Analysis of a multiplicative hybrid route choice model in stochastic assignment paradox", *Transportmetrica A: Transport Science*, pp. 1–25, 2021. DOI: 10.1080/23249935.2021.1953189

[23] X. Wang, **Z. Cheng**, M. Trépanier, and L. Sun*, "Modeling bike-sharing demand using a regression model with spatially varying coefficients", *Journal of Transport Geography*, vol. 93, p. 103 059, 2021. DOI: 10.1016/j.jtrangeo.2021.103059

[24] J. Yao*, **Z. Cheng**, J. Dai, A. Chen, and S. An, "Traffic assignment paradox incorporating congestion and stochastic perceived error simultaneously", *Transportmetrica A: Transport Science*, vol. 15, no. 2, pp. 307–325, 2019. DOI: 10.1080/23249935.2018.1474962

[25] J. Yao*, W. Huang, A. Chen, **Z. Cheng**, S. An, and G. Xu, "Paradox links can improve system efficiency: An illustration in traffic assignment problem", *Transportation Research Part B: Methodological*, vol. 129, pp. 35–49, 2019. DOI: 10.1016/j.trb.2019.07.018

[26] J. Yao*, **Z. Cheng**, F. Shi, S. An, and J. Wang, "Evaluation of exclusive bus lanes in a tri-modal road network incorporating carpooling behavior", *Transport Policy*, vol. 68, pp. 130–141, 2018. DOI: 10.1016/j.tranpol.2018.05.001

Preprints & Working papers

[1] Y. Zhou, **Z. Cheng**, L. Hu, Y. Bu, and S. Wang*, "Nestgnn: A graph neural network framework generalizing the nested logit model for travel mode choice", 2025. arXiv: 2509.07123.

[2] Y. Wu, **Z. Cheng**, and L. Sun, "Individual mobility prediction via attentive marked temporal point processes", 2021. arXiv: 2109.02715.

Awards and Scholarships

- Best Paper Award at Transportation Research Board (TRB) Statistical Methods Committee 2024

- Second-best Paper Award at the 15th CASPT and the 8th TransitData. 2022
- CIRRELT Excellence Scholarship (Doctoral Rédaction) 2020
- McGill Engineering Doctoral Award (International) 2019–2021
- Excellent Graduate Thesis of HIT 2018
- First Level Scholarship of HIT 2016, 2017
- Excellent Graduate of Shandong Province 2016
- Third Prize of National Competition of Transport Science and Technology 2015
- First Prize of China Undergraduate Mathematical Contest in Modeling (team leader) 2014

Teaching and Department Service

- **Coordinator** 2024-2025
  Master of Science program in Urban Analytics at the University of Florida
- **Lecturer** Fall 2023
  Spatiotemporal Data Mining (CIVE 650) at McGill University
- **Guest Lecturer** Winter 2023, Winter 2024
  Transportation Network Analysis (CIVE 542) at McGill University
- **Teaching Assistant** Fall 2022
  Spatiotemporal Data Mining (CIVE 650) at McGill University
- **Teaching Assistant** Winter 2021, Fall 2021
  Traffic Engineering & Simulation (CIVE 440) at McGill University
- **Teaching Assistant** Winter 2021
  Sustainable project management (CIVE 324) at McGill University

Students mentoring

Co-supervising

- Ben Chi (Ph.D. student at Zhejiang University) June 2026–Current
  - Topic: Large Language Model for mobility prediction and generation
- Zhaohao Mei (Ph.D. student at Zhejiang University) June 2026–Current
  - Topic: Mobility demand prediction

Assist in mentoring

- Yuqi Zhou (Ph.D. student at the University of Florida) 2024–2026
  - Topic: Deep-learning-based choice modeling for travel mode analysis
- Xiaoxu Chen (Ph.D. student at McGill University) 2021–2024
  - Topic: Probabilistic forecasting for bus travel time and passenger occupancy
- Zixu Zhuang (Undergraduate at Harbin Institute of Technology) 2017–2018
  - Topic: Bus travel time reliability

Projects and Grants

- 生成式交通出行预测与优化 (Generative Mobility Forecasting and Optimization) Jan 2027–Dec 2029
  - National Natural Science Foundation of China
  - Role: Principal Investigator (PI)
- "Hundred Talents Program" Professor Start-up Funding Jun 2026–Jun 2032

  - Zhejiang University
  - Role: Principal Investigator (PI)
- Adaptive Mobility Services for Low-Income Health Care Workers and Patients through a Multi-level Connected and Automated Transit System (M-CATS) Aug 2024–Dec 2025
  - U.S. DOE-funded project with team members from MIT, University of Florida, Northeastern University, Chicago Transit Authority, Illinois Medical District, and other industry partners. USD 3+ million
  - Role: Participant
- Deep Probabilistic Demand Forecasting for Emerging Mobility Systems Jan 2024–Jan 2025
  - NSERC Alliance International Catalyst Grant with TU Delft CAD 25,000
  - Role: Participant
- Probabilistic Forecasting of Train Ticket Booking Demand with Hierarchical Correlations Aug 2022–Aug 2024
  - NSERC Alliance project with Mitacs and ExPretio Inc. CAD 240,000
  - Role: Investigator (contributed significantly to grant writing)
- Enhancing Transit Service by Intelligent Trip Inference and Recommendation System Nov 2021–Aug 2022
  - NSERC Alliance project with Transit.app CAD 30,000
  - Role: Investigator (contributed significantly to grant writing)
- Spatiotemporal Travel Behavior Modeling and Analysis for Better Public Transport Systems 2019–2022
  - Mitacs Accelerate project with exo CAD 160,000
  - Role: Investigator (contributed significantly to grant writing)
- Research on Spatiotemporal Characteristics of Travel Route Selection Based on Big Data 2018–2019
  - GAIA collaborative research funds for young scholars with DiDi Global Inc. RMB 150,000
  - Role: Participant
- Research on the Characteristics of Traffic Paradox in Random Route Choice Model 2016–2018
  - National Natural Science Foundation of China RMB 207,200
  - Role: Participant

INVITED TALKS

[1] "The Fourth Paradigm: AI & Data-driven Intelligent Transportation", The University of New South Wales (UNSW), Syndey, Australia (online), Nov. 5, 2025.

[2] "Modeling Residential Location Choice with Graph Neural Networks", Urban Mobility Lab at Massachusetts Institute of Technology (MIT), Dec. 11, 2024.

[3] "Travel behavior for urban mobility prediction", SERMOS Lab and JGT Lab at University of Florida, Nov. 1, 2024.

[4] "Data-driven + Transportation Science: New Ideas for Urban Mobility Modeling", Southeast University, International Young Scholars Forum, Naijing, China (online), Dec. 27, 2023.

[5] "Deep probabilistic forecasting of zero-inflated count data", ExPretio, Montreal, Canada, Oct. 31, 2023.

[6] "The regularity, predictability, and travel behavior in urban transit mobility", University of Toronto Institute of Transportation Engineers (UT-ITE) Student Chapter, Toronto, Canada, Oct. 27, 2023.

[7] "Real-time forecasting of metro origin-destination matrices with high-order weighted dynamic mode decomposition", ExPretio, Montreal, Canada, Dec. 15, 2022.

[8] "Travel-behavior-based inference and forecasting methods in metro systems", School of Economics and Management, Dalian University of Technology, Dalian, China (online), Nov. 26, 2022.

[9] "Probabilistic model for destination inference and travel pattern mining from smart card data", Zooming in on collaborative digital intelligence, Montreal, Canada (online), Apr. 21, 2021. [Online]. Available: https://youtu.be/xLuYrb_mmdM

Professional services

Editor

- Guest editor of the special issue "Innovative Applications of Operations Research and Machine Learning in Traffic and Transportation Management" in Multimodal Transportation

Journal Reviewer

- Transportation Research Part A: Policy and Practice
- Transportation Research Part B: Methodological
- Transportation Research Part C: Emerging Technologies
- Transportation Research Part E: Logistics and Transportation Review
- Transportation Science
- INFORMS Journal on Computing
- IEEE Transactions on Intelligent Transportation Systems
- Transport Policy
- Transportmetrica A: Transport Science
- Artificial Intelligence for Transportation
- Transportation Research Record
- Public Transport
- Journal of Advanced Transportation
- Urban Studies
- Journal of Planning Education and Research
- Sustainability
- International Journal of Intelligent Systems

Conference Reviewer

- IEEE International Conference on Intelligent Transportation Systems (ITSC)
- TRISTAN 2022 (The 11th Triennial Symposium on Transportation Analysis conference)
- Transportation Research Board (TRB) Annual Meeting
- COTA International Conference of Transportation Professionals

Membership

- Youth Member, Technical Committee of Transportation Network Analysis and Optimization, World Transport Convention, December 2023 – October 2026.
- Member of IEEE, 2022–Current
- Student member of Chinese Overseas Transportation Association (COTA) 2020–2022
- Student member of Interuniversity Research Centre on Enterprise Networks, Logistics and Transportation (CIRRELT), 2019–2022
- Friend member of TRB Standing Committee on Urban Transportation Data and Information Systems (AED20), 2021–Current
- Friend member of TRB Standing Committee on Public Transport Planning and Development (AP025), 2021–Current
- Student member of McGill Sustainability Systems Initiative (MSSI), 2021–2022

Conferences

[1] "Graph Neural Networks for Residential Location Choice: Connection to Classical Logit Models", in *Emerging Challenges in Statistical Modeling for Transportation Research Workshop at the Banff International Research Station (BIRS)*, Banff, Canada, Jun. 30, 2026.

[2] Y. Zhou, **Z. Cheng**, L. Hu, Y. Bu, and S. Wang, "Nestgnn: A graph neural network framework generalizing the nested logit model for travel mode choice", in *Transportation Research Board 105th Annual Meeting*, Washington, D.C., 2026.

[3] X. Chen, **Z. Cheng**, and L. Sun, "Probabilistic forecasting of bus travel time with a bayesian hidden markov mesh model", in *Transportation Research Board 104th Annual Meeting*, Washington, D.C., 2025.

[4] **Z. Cheng**, J. Wang, M. Trépanier, and L. Sun, "Abnormal metro passenger demand is predictable from alighting and boarding correlation", in *Transportation Research Board 104th Annual Meeting*, Washington, D.C., 2025.

[5] X. Chen, **Z. Cheng**, A. M. Schmidt, and L. Sun, "Probabilistic forecasting of bus travel time and passenger occupancy with bayesian time-dependent continuous density hidden markov model", in *Transportation Research Board 103th Annual Meeting*, Washington, D.C., 2024.

[6] X. Chen, **Z. Cheng**, C. Zhang, L. Sun, and S. Nicolas, "Memory-efficient hankel tensor factorization for extreme missing traffic data imputation", in *Transportation Research Board 103th Annual Meeting*, Washington, D.C., 2024.

[7] **Z. Cheng**, J. Wang, M. Trépanier, and L. Sun, "Anomalies in metro passenger demand are predictable – learning causality with ABtransformer", in *TransitData 2024*, London, 2024.

[8] F. Wu, **Z. Cheng**, H. Chen, T. Z. Qiu, and L. Sun, "Traffic state estimation from vehicle trajectories with anisotropic gaussian processes", in *Transportation Research Board 103th Annual Meeting*, Washington, D.C., 2024.

[9] H. Cai, F. Wu, **Z. Cheng**, B. Li, and J. Wang, "Study on impact factors of charging station utilization using random forest and shap value methods", in *Transportation Research Board 102th Annual Meeting*, Washington, D.C., 2023.

[10] X. Chen, **Z. Cheng**, S. Nicolas, and L. Sun, "Laplacian convolutional representation for traffic time series imputation", in *World Conference on Transport Research (WCTR)*, Montreal, 2023.

[11] **Z. Cheng**, X. Wang, X. Chen, M. Trépanier, and L. Sun, "Bayesian calibration of traffic flow fundamental diagrams using gaussian processes", in *Transportation Research Board 102th Annual Meeting*, Washington, D.C., 2023.

[12] F. Wu, **Z. Cheng**, and L. Sun, "Traffic state imputation with gaussian processes", in *World Conference on Transport Research (WCTR)*, Montreal, 2023.

[13] X. Chen, **Z. Cheng**, and L. Sun, "Bayesian inference for link travel time correlation of a bus route", in *Conference on Advanced Systems in Public Transport (CASPT) and TransitData 2022*, Tel Aviv, 2022.

[14] **Z. Cheng**, M. Trépanier, and L. Sun, "Real-time forecasting of metro origin-destination matrices with high-order weighted dynamic mode decomposition", in *Conference on Advanced Systems in Public Transport (CASPT) and TransitData 2022*, Tel Aviv, 2022.

[15] X. Wang, **Z. Cheng**, M. Trépanier, and L. Sun, "Modeling bike-sharing demand using a regression model with spatially varying coefficients", in *Transportation Research Board 100th Annual Meeting*, Washington, D.C. (virtual), 2021.

[16] **Z. Cheng**, H. Alizadeh, M. Nazem, M. Trépanier, and L. Sun, "Long-term ridership forecast using heuristic, SARIMA and random forest methods", in *TransitData 2020*, Toronto (virtual), 2020.

[17] **Z. Cheng**, M. Trépanier, and L. Sun, "Integrating travel behavior regularity into passenger flow prediction", in *TransitData 2020*, Toronto (virtual), 2020.

[18] **Z. Cheng**, M. Trépanier, and L. Sun, "Inferring trip destinations in transit smart card data using a probabilistic topic model", in *TransitData 2019*, Paris, 2019.

[19] Z. Zhuang, **Z. Cheng**, J. Yao, J. Wang, and S. An, "Bus travel time reliability incorporating in-stop waiting time and in-vehicle travel time with AVL data", in *Transportation Research Board 98th Annual Meeting*, Washington, D.C., 2019.

[20] J. Yao, **Z. Cheng**, S. An, and A. Chen, "Analysis of a multiplicative hybrid route choice model in stochastic assignment paradox", in *Transportation Research Board 97th Annual Meeting*, Washington, D.C., 2018.

[21] J. Yao, J. Dai, A. Chen, **Z. Cheng**, and S. An, "Traffic assignment paradox incorporating congestion and stochastic perceived error simultaneously", in *Transportation Research Board 97th Annual Meeting*, Washington, D.C., 2018.